

Curriculum Bot – Intent (Question Category)

Description

Service Name: curriculum_bot

This document describes the categories of questions the Curriculum Bot can answer using the current YAML configuration and table documentation present in the repository.

Scope limitation (important):

- Entity types used for value resolution are defined only in `curriculum-agent/backend/docs/entities/entities.yaml`.
- Table-level schema grounding is driven only by table documentation YAMLs in `curriculum-agent/backend/docs/tables/*.yaml`.
- In the current repository snapshot, only `aca_batch.yaml` exists as a concrete table YAML, so the bot can reliably answer questions about the fields defined in `aca_batch`.

0. Shared filtering rules (applies to all categories)

Global domain rules from `backend/docs/rules/global_rules.yaml`:

- Default record filtering:
 - Always filter records by `record_status = 'A'` unless the user explicitly asks for inactive or deleted records.
- Batch year meaning:
 - `batch_year` refers to the student admission year (e.g., "2023 batch" = students who joined in 2023).
- Counting behavior:
 - When counting programmes or batches, always use `DISTINCT` on `programme_name` and `campus_name` to avoid duplicates.
- Join behavior:
 - Never assume a JOIN path that is not listed in the provided `JOIN_PATHS` (the bot can only join tables when join paths exist).

Batch year interaction (online agent behavior):

- If the user question does not specify a batch year, the agent will ask the user to choose:

- "Latest Batch Year (current_year - 1)"
- "All Batches (No specific year)"

1. Campus Information (Campus identity and queries by campus)

What the bot answers:

- Questions asking for campus identity, campus code/name, or campus-related attributes for a particular batch.

YAML-backed entity / value resolution:

- Entity type: `campus_name`
 - Canonical column: `aca_batch.campus_name`
 - Alias column(s): `aca_batch.campus_short_name`
 - Extraction filter: entity config includes `filter_condition: record_status = 'A'`

`aca_batch.yaml` fields used for these questions:

- `campus_name`
- `campus_short_name`
- `location_name`
- `location_short_name`

Typical question patterns:

- "Which campus offers ... for 2024 batch?"
- "Show programmes in BCC campus"
- "What is the location for programme ... at Bangalore Central Campus?"

2. Department Information (Department identity and department-level queries)

What the bot answers:

- Questions asking for department identity and department-level attributes for a particular batch.

YAML-backed entity / value resolution:

- Entity type: `department_name`

- Canonical column: `aca_batch.department_name`
- Alias column(s): `aca_batch.department_code`
- Extraction filter: `record_status = 'A'`

`aca_batch.yaml` fields used for these questions:

- `department_name`
- `department_code`
- `department_category_name`
- `is_category_academic`
- `is_current_session` (when users ask "current" campus/department programmes)

Typical question patterns:

- "Show programmes offered by the Psychology department in 2024 batch"
- "Which departments have current active programmes?"
- "List department codes for active programmes"

3. Programme Degree (Degree-level programme queries)

What the bot answers:

- Questions that ask for programmes filtered by degree (e.g., Master of Science, PhD, etc.) and batch scoping.

YAML-backed entity / value resolution:

- Entity type: `programme_degree`
 - Canonical column: `aca_batch.programme_degree`
 - Alias columns: none defined in `entities.yaml` for this entity type (currently only the name column is used)
 - Extraction filter: `record_status = 'A'`

`aca_batch.yaml` fields used for these questions:

- `programme_degree`
- `programme_level`
- `programme_commence_year`
- `programme_inactivated_year`
- `record_status`

Typical question patterns:

- "What programmes exist for Master of Science in latest batch?"
- "Which PhD degrees are active in 2023 batch?"

4. Programme Name (Programme-level queries by exact program name)

What the bot answers:

- Questions asking for programme identity (name/code/short name) and all programme attributes for a specified batch.

YAML-backed entity / value resolution:

- Entity type: `programme_name`
 - Canonical column: `aca_batch.programme_name`
 - Alias column(s): `aca_batch.programme_code`,
`aca_batch.programme_short_name`
 - Extraction filter: `record_status = 'A'`

`aca_batch.yaml` fields used for these questions:

- `programme_name`
- `programme_code`
- `programme_short_name`
- `programme_degree`
- `programme_level`
- `mode`
- `program_objectives`
- `introduction_to_the_program`
- `adm_programme_selection_alert_text`
- `programmes_combined`
- `is_specialization`

Typical question patterns:

- "Give me details for 'Master of Science (...)' in 2024 batch"
- "Show programme objectives for programmes offered by ... at ... in latest batch"

5. Batch Identity and Intake Scope (batch_name, academic year, and batch identifiers)

What the bot answers:

- Questions that focus on the specific batch record(s): batch naming, admissions year, academic year label, and intake numbering.

YAML-backed filtering behavior:

- `batch_year` is detected/clarified by the agent and is interpreted as admission year.
- Global filtering defaults to `record_status = 'A'`.

`aca_batch.yaml` fields used for these questions:

- `aca_batch_id`
- `batch_name`
- `batch_year`
- `batch_year_name`
- `academic_year`
- `academic_year_name`
- `batch_commencement_date`
- `batch_commencement_month`
- `programme_completion_month`
- `admission_cancellation_till_date`
- `in_take_batch_number`
- `number_of_batches`
- `total_session_intakes_in_year`
- `total_sessions_in_year`

Typical question patterns:

- "List active batches for 2024 batch"
- "What is the academic_year_name for batch ...?"
- "How many intakes exist in a given academic year?"

6. Admission and Eligibility (intake capacity and eligibility text)

What the bot answers:

- Questions about admission eligibility and approved/revised intake capacities.

`aca_batch.yaml` fields used for these questions:

- `program_admission_eligibility`
- `approved_intake_for_batch`
- `revised_intake_for_batch`
- `admission_cancellation_till_date`
- `is_aided_programme`

Typical question patterns:

- "What is the approved intake for Programme X in 2024 batch?"
- "What is the revised intake for a programme at campus Y?"
- "What are the admission eligibility requirements for ... in latest batch?"

7. Programme Duration and Graduation Requirements (duration, credits, and graduation minima)

What the bot answers:

- Questions about how long programmes take and what minimum requirements are needed for graduation.

`aca_batch.yaml` fields used for these questions:

- `regular_duration_in_years`
- `number_of_regular_semesters`
- `min_no_of_years_for_graduation`
- `max_years_to_graduate`
- `min_no_of_credits_for_graduation`
- `min_core_courses`
- `min_major_credit_required`
- `min_elective_courses`
- `min_minor_credit_required`

Typical question patterns:

- "What is the minimum number of credits for graduation in Programme X?"
- "How many semesters are in the regular programme duration?"

- "What are the graduation time limits for latest batch of ...?"

8. Honours, Dissertation, and Research Track Flags

What the bot answers:

- Questions about honours applicability and associated research/dissertation requirements for a programme/batch.

`aca_batch.yaml` fields used for these questions:

- `is_honours_programme`
- `is_honours_programmes_with_research`
- `is_dissertation_required`
- `honours_applicable_year`
- `honours_duration_in_years`
- `number_of_honours_semesters`

Typical question patterns:

- "Show honours programmes with research in 2023 batch"
- "Is a dissertation required for honours track of Programme X?"

9. Internship, Field Projects, and Add-ons (implementation flags)

What the bot answers:

- Questions about whether internship/field project/add-on courses apply to the programme/batch.

`aca_batch.yaml` fields used for these questions:

- `is_internship_applicable`
- `is_field_project_applicable`
- `is_addon_course_applicable`

Typical question patterns:

- "Are internships applicable for Programme X in 2024 batch?"
- "Is a field project mandatory for ...?"
- "Are add-on courses applicable for latest batch of Programme Y?"

10. Curriculum Frameworks and Policy Flags (CBCS/ECS/NEP and related curriculum structure)

What the bot answers:

- Questions about whether CBCS/ECS/NEP are implemented and related curriculum configuration flags.

`aca_batch.yaml` fields used for these questions:

- `is_nep_programme`
- `nep_initiated_year`
- `is_implemented_CBCS`
- `CBCS_implemented_year`
- `is_implemented_ECS`
- `ECS_implemented_year`
- `curriculum_completion_type`
- `batch_creation_type`

Typical question patterns:

- "Is NEP implemented for programme ... in 2024 batch?"
- "When was CBCS implemented for a programme?"
- "What completion type does Programme X follow?"

11. Academic Structure Constraints (elective limits, specialization, majors/exits)

What the bot answers:

- Questions that involve major/minor/elective structure limits, multi-major, and exit constraints.

`aca_batch.yaml` fields used for these questions:

- `is_multiple_major`
- `is_multiple_exit`
- `is_having_pso`
- `is_having_peo`
- `max_elective_from_same_disciplinary`

- `max_elective_from_other_disciplinary`
- `is_second_language_included`
- `programmes_combined`
- `is_specialization`

Typical question patterns:

- "What elective limits apply for Programme X?"
- "Is second language included in the programme curriculum?"
- "Is it a specialization programme and does it support multiple majors?"

12. Academic Calendar and Session Dates (session timeline, exams, elective windows, course planning)

What the bot answers:

- Questions about session dates, exam windows, elective registration windows, and course planning timelines.

`aca_batch.yaml` fields used for these questions:

- `duration_name`
- `session_type_name`
- `session_start_date`
- `session_end_date`
- `session_first_instruction_date`
- `session_last_instruction_date`
- `session_final_exam_start_date`
- `session_final_exam_end_date`
- `is_current_session`
- `academic_duration_start_date`
- `academic_duration_end_date`
- `open_elective_start_date`
- `open_elective_end_date`
- `open_elective_registration_start_date`
- `open_elective_registration_end_date`
- `course_plan_start_date`
- `course_plan_end_date`
- `vacation_start_date`

- `vacation_end_date`

Typical question patterns:

- "When do final exams start for Programme X in 2024 batch?"
- "What are the open elective registration dates?"
- "Is the session currently active?"

13. Status, Sync, and Lifecycle (active/inactive, programme activation year, synchronization)

What the bot answers:

- Questions explicitly about status, inactive/inactivated periods, and sync timestamps.

Global default:

- The bot defaults to `record_status = 'A'` unless the user explicitly asks otherwise.

`aca_batch.yaml` fields used for these questions:

- `record_status`
- `sync_datetime`
- `programme_commence_year`
- `programme_inactivated_year`
- `is_change_in_department`
- `programme_established_year`
- `sap_code`

Typical question patterns:

- "What changed in department ownership for Programme X?"
- "Show inactive programmes for a given batch year."
- "When was the record last synced?"

14. Aggregation and Counting Questions (how many distinct programmes/batches)

What the bot answers:

- Questions requesting counts/aggregations over programmes/batches.

YAML-backed rule:

- Use `DISTINCT` on `programme_name` and `campus_name` when counting programmes/batches to avoid duplicates.

How this maps to `aca_batch.yaml` fields:

- Counting is performed over rows using fields like:
 - `programme_name`
 - `campus_name`
 - `batch_year` (when the question scopes by admission year)
 - `record_status` (active by default)

Typical question patterns:

- "How many active programmes are offered in 2024 batch?"
- "How many programmes are in Computer Science department for latest batch?"
- "Count distinct programme_name + campus_name combinations."